

Decoding cortical layer from functional fingerprints in balanced MICrONS V1 excitatory neurons

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1 Introduction

Understanding how neuronal circuits encode visual information requires linking neural activity to circuit structure. The MICrONS dataset provides an unusually rich setting for this problem because it includes both neural activity recorded while the mouse viewed visual stimuli and anatomical information such as cell identity, cortical layer, and synaptic connectivity.

Functional imaging studies have shown that visual responses can be measured and compared across cortical layers, including deep layers in awake mice. More recently, large-scale models of neural activity have suggested that learned functional representations can contain information about anatomical properties, including cell type and structural features. Our project tests a simpler and more controlled version of this idea: whether directly measured response patterns to a shared visual stimulus set are sufficient to recover cortical layer.

2 Research question

The main research question is:

Can the cortical layer of a V1 excitatory neuron be decoded from its functional responses?

More specifically, we wanted to understand which representation of the functional response carries the most useful layer information. Is the signal present in simple response magnitude? In response reliability? In behavioural or stimulus covariates? In the pattern of responses across stimuli? Or in the raw temporal shape of the calcium trace?

3 Methodology

3.1 Dataset selection

We analysed V1 excitatory neurons from the MICrONS functional-connectomics dataset. The supervised target was cortical layer, with four classes: L2/3, L4, L5 and L6. Layer 1 was excluded because it is severely underrepresented among excitatory neurons and is not comparable to the other layers for this decoding task.

Starting from 8,895 V1 excitatory best-only matched neurons, we first applied a label-consistency filter. Each AIBS excitatory cell-type label was mapped to its expected layer.

Neurons whose cell-type-implied layer disagreed with their depth-derived layer label were removed. This excluded 407 neurons and left 8,488 label-consistent neurons. We then selected session-scans containing at least 20 neurons in each of the four target layers. This produced the final controlled benchmark: five balanced sessions (5.6, 5.7, 6.2, 6.4, 6.6), containing 3,545 neurons in total.

The final layer counts were: $L_{2/3} = 1456$, $L_4 = 959$, $L_5 = 841$, $L_6 = 289$. This restriction was chosen to make the comparison between functional representations fair and interpretable.

3.2 Functional preprocessing and representations

All main analyses were restricted to the 116 oracle stimulus hashes, i.e. visual stimuli repeated across all selected sessions. This ensured that every neuron was represented using the same stimulus set and that model comparisons were not confounded by differences in stimulus availability. Repeated trials of the same stimulus were averaged within each neuron-hash pair.

We compared three functional representations. The first representation used engineered tabular features extracted from each trial-averaged calcium trace. These included neural response amplitude and shape features (A1), trial-by-trial reliability features (B), behavioural covariates such as running and pupil variables (C1), and stimulus descriptors (D1). All engineered scalar features reported in the main analysis were z-scored within session to reduce scan-level amplitude offsets caused by imaging depth, gain, optical quality or expression differences.

The second representation was the oracle response fingerprint, Tier G. Each neuron was represented by its response vector to the 116 shared oracle stimuli:

$$x_n = [r_{n,h_1}, r_{n,h_2}, \dots, r_{n,h_{116}}]$$

where $r_{n,h}$ is the trial-averaged response amplitude of neuron n to stimulus h . This representation preserves stimulus identity and tests whether layer information is carried by the neuron's pattern of visual selectivity.

The third representation used raw trial-averaged calcium traces as input to a 1D CNN. The CNN tested whether temporal response shape contains information beyond scalar summaries or stimulus-specific response magnitude.

3.3 Models and neuron-level prediction

We compared logistic regression, Histogram Gradient Boosting (HGB), Deep Sets and CNN models. Logistic regression served as a linear baseline. HGB was used as a strong nonlinear tabular model. Deep Sets treated each neuron as an unordered set of stimulus responses. CNNs encoded raw calcium traces and pooled information across oracle stimuli.

A central methodological point is that the biological label is neuron-level, while tabular observations are neuron-stimulus pairs. Therefore, the final prediction must be one layer label per neuron. For classical tabular models, the classifier produced a probability vector for each neuron-hash row, and probabilities were averaged across the 116 hashes of the same neuron. For Deep Sets and CNNs, aggregation was performed inside the model: hash-level responses or traces were encoded, pooled across stimuli, and classified at the neuron level.

3.4 Evaluation and controls

The primary metric was balanced accuracy,

$$BA = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c$$

with $C = 4$ layers. Balanced accuracy was used because the classes were still imbalanced. The four-class chance level is therefore $BA = 0.25$.

The main generalization test was leave-one-session-out cross-validation (LOSO). In each fold, the model was trained on four sessions and tested on the held-out fifth session. This evaluates whether the learned relationship between function and layer transfers to a recording session not seen during training.

To test whether performance was driven by confounds, we included scan-only and `cc_abs`-only baselines. The scan-only baseline used only session identity and tests whether layer can be predicted from session composition. The `cc_abs` baseline used only a response-quality scalar and tests whether signal quality alone explains decoding performance.

4 Results

4.1 Engineered tabular features

The first analysis tested whether interpretable scalar summaries of the calcium response were sufficient to decode cortical layer. The model comparison showed that representation geometry matters. Logistic regression was consistently weaker and partially collapsed on some classes, especially L2/3, indicating that the layer signal is not well captured by a simple linear boundary in the engineered feature space. HGB improved over logistic regression, consistent with nonlinear interactions among response features. Deep Sets performed best among the tabular models, reaching:

$$BA_{LOSO} = 0.52 \pm 0.02$$

with the full $A1 + B + C1 + D1$ representation. This supports the methodological premise that a neuron is more naturally represented as a set of responses to many stimuli rather than as independent neuron-stimulus rows.

Comparing this value with the result obtained with the neural-response feature block only (A1), which yield $BA = 0.49$, we see that the improvement from adding reliability, behavioural variables and stimulus descriptors is limited.

4.2 Oracle response fingerprint

The headline model was HGB on the z-scored Tier G fingerprint (see Fig.1). Across five random seeds, it achieved:

$$BA_{LOSO} = 0.545 \pm 0.008$$

This was the strongest and most stable main result. The small seed variability indicates that the result was not due to a lucky initialization or stochastic training artifact. The fingerprint result was also well above the confound baselines. The scan-only baseline was exactly at chance, while the response-quality baseline, `cc_abs`-only, performed above chance but clearly below Tier G:

$$BA_{scan-only} = 0.25 \quad BA_{cc_abs-only} = 0.38$$

This means that response quality or reliability contains some layer-associated information, but it does not explain the full fingerprint result. The model’s performance requires more than a scalar measure of recording quality or response consistency.

4.3 CNN temporal controls

The CNN experiment tested whether raw calcium trace shape contains additional layer information beyond engineered summaries and oracle response fingerprints. A 1D CNN can in

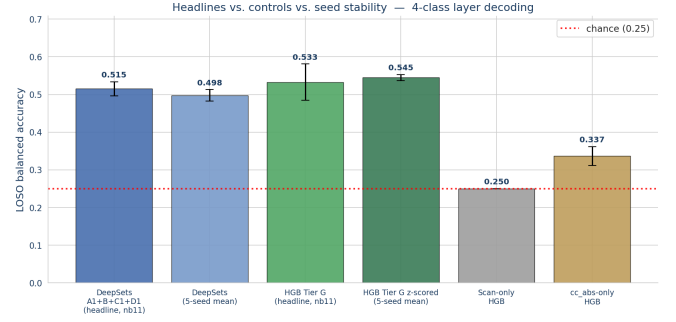


Figure 1: Oracle fingerprint.

principle exploit temporal motifs such as response onset, peak timing, decay, transient activity or sustained activation. Therefore, the relevant question was not only whether the CNN performed well, but whether it depended on the temporal order of the trace. The intact hash-aware CNN with attention pooling reached:

$$BA_{intact} = 0.535 \pm 0.047.$$

This placed CNN performance in the same range as the Tier G fingerprint. However, the temporal controls changed the interpretation. The time-shuffled CNN, which destroys temporal order while preserving the distribution of trace values, reached:

$$BA_{time-shuffled} = 0.547 \pm 0.051.$$

Because the time-shuffled model was not worse than the intact model, these results do not support the conclusion that fine temporal order in the calcium trace is the main source of layer information.

5 Discussion

The results answer the main research question positively, but only partially: cortical layer is decodable from functional responses above chance, yet the decoding is far from perfect. This suggests that functional activity contains information about laminar identity, but does not fully determine it. The strongest evidence comes from the oracle response fingerprint, indicating that the relevant signal is not simply average response magnitude, reliability, or generic temporal shape, but the pattern of responses across specific visual stimuli.

Several limitations constrain the interpretation. The main benchmark used five balanced, label-consistent sessions, which makes model comparison cleaner but limits the claim relative to the full MICRONS dataset. Calcium imaging is an indirect and temporally smoothed proxy for spiking, so the CNN control should not be read as evidence that precise neural timing is biologically irrelevant. Finally, L6 remains underrepresented and layer labels can be ambiguous near boundaries.

Alternative analyses can broaden this benchmark by including all sessions with stronger imbalance correction, using hierarchical models that first decode layer and then subtype, or learning embeddings from larger neural populations. The exploratory all-session, stimulus-family and subtype analyses are first steps in this direction; future work should validate the result across animals, identify the specific visual features driving the fingerprint, and test whether fine excitatory subtype identity is recoverable beyond the layer signal.

Appendix

A Dataset composition and label consistency

This appendix justifies the controlled five-session population used in the main analysis. Each AIBS excitatory cell-type label was mapped to its expected cortical layer, and neurons whose cell-type-implied layer disagreed with their depth-derived layer label were treated as mismatched. Across the full population, 407 of 8,895 neurons (3.85%) were mismatched. Mismatches are not uniformly distributed: they concentrate in the deep layers (L5 11.8%, L6 16.3%) and in a small number of session-6 scans (Fig. 2). For mismatched neurons, the model trained on depth-derived labels mostly follows the depth label rather than the cell-type label (Fig. 3), and per-target accuracy on mismatched neurons is lower than on matched ones; consistent with these being genuine boundary cells or annotation ambiguities rather than a systematic bias. All mismatched neurons were removed before session selection.

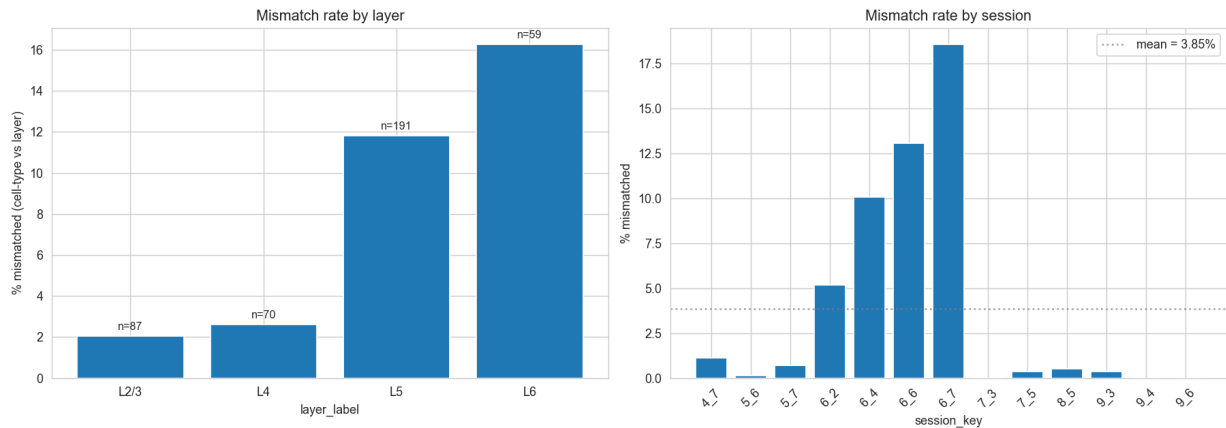


Figure 2: Label-mismatch rate by layer and by session. Cell-type vs depth-derived layer disagreement is concentrated in deep layers and in a few session-6 scans.

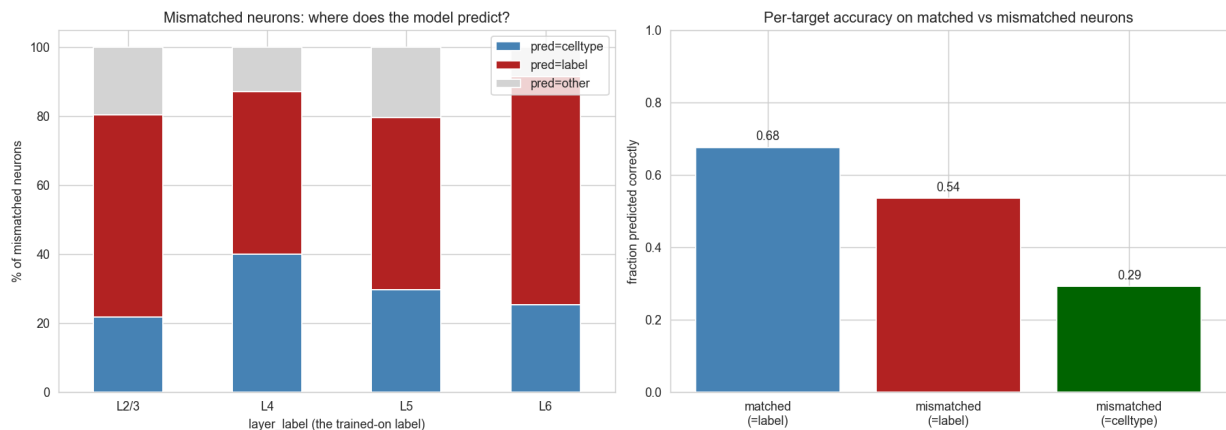


Figure 3: Where the model predicts for mismatched neurons. Left: for mismatched neurons the model mostly predicts the depth-derived label, not the cell-type label. Right: per-target accuracy is lower on mismatched than on matched neurons.

Table 1: Full session composition after label-consistency filtering. Per-session, per-layer neuron counts across all 13 V1 scans, after removing the 407 cell-type/layer mismatches. “Well-behaved” marks the scans with ≥ 20 neurons in every target layer, which form the main benchmark.

Session	L2/3	L4	L5	L6	Total	Well-behaved
4_7	364	240	103	0	707	no
5_6	151	249	189	96	685	yes
5_7	184	284	147	81	696	yes
6_2	268	280	153	33	734	yes
6_4	452	122	200	50	824	yes
6_6	401	24	152	29	606	yes
6_7	453	52	192	15	712	no
7_3	203	190	159	0	552	no
7_5	176	76	21	0	273	no
8_5	400	268	108	0	776	no
9_3	0	815	0	0	815	no
9_4	680	0	0	0	680	no
9_6	428	0	0	0	428	no
Total	4160	2600	1424	304	8488	

Table 2: Final five-session benchmark counts. The well-behaved subset used for all main representation comparisons: 3,545 label-consistent V1 excitatory neurons.

Session	L2/3	L4	L5	L6	Total
5_6	151	249	189	96	685
5_7	184	284	147	81	696
6_2	268	280	153	33	734
6_4	452	122	200	50	824
6_6	401	24	152	29	606
Total	1456	959	841	289	3545

B Tabular feature description

Tier A1: Neural response amplitude and temporal shape

Tier A1 contains 20 features extracted from the trial-averaged calcium response trace for a given neuron and stimulus hash. These features describe the response itself: how large it is, how variable it is, when it peaks, and how it evolves over time.

A-amplitude features

The first 10 features summarize response amplitude and global trace statistics.

Feature	Description
amp_baseline_mean	Mean activity during the first second of the trace, used as a pre-stimulus or early baseline estimate.
amp_baseline_std	Standard deviation of the baseline period, measuring baseline variability.
amp_mean	Mean value of the entire calcium trace.
amp_peak	Maximum value of the trace, representing the strongest response amplitude.
amp_min	Minimum value of the trace; often close to zero for deconvolved or non-negative responses.
amp_range	Difference between the peak response and the baseline mean: $\text{amp_range} = \text{amp_peak} - \text{amp_baseline_mean}.$
amp_integral	It measures response amplitude above baseline. Sum of the trace values, approximately corresponding to the area under the curve (AUC). It captures total response magnitude over time.
amp_tail_mean	Mean activity during the last second of the trace, measuring the late or post-stimulus tail of the response.
amp_variance	Variance of the full trace.
amp_std	Standard deviation of the full trace.

A-shape features

The second 10 features summarize the temporal shape of the trial-averaged response.

Feature	Description
shape_latency_s	Time of the response peak in seconds. It measures how early or late the neuron reaches its maximum response.
shape_peak_frame_idx	Frame index of the response peak; this is the discrete-frame version of peak latency.
shape_rise_slope	Slope of the response rise, approximated as: $\frac{\text{peak} - \text{baseline}}{\text{latency}}.$
shape_decay_slope	It measures how quickly the response increases toward the peak. Slope of the response decay, approximated as: $\frac{\text{peak} - \text{tail_mean}}{\text{duration} - \text{latency}}.$
shape_early_auc	It measures how quickly the response decreases after the peak.
shape_late_auc	Sum of trace values in the first half of the response window.
shape_early_late_ratio	Sum of trace values in the second half of the response window. Ratio between early and late response area: $\frac{\text{early_auc}}{\text{late_auc}}.$
shape_adaptation_ratio	It measures whether the response is concentrated early or late. Ratio between late and early activity, computed using the last and first quartiles of the trace. It captures adaptation or persistence of the response.
shape_center_of_mass_s	Temporal center of mass of the response, indicating whether the response is concentrated early or late in time.
shape_width_above_half_max_s	Duration for which the trace remains above half of its maximum value. This measures response width or persistence.

Tier B: Trial-by-trial reliability

Tier B contains 7 features that describe how reliable or reproducible the response is across repeated trials of the same oracle stimulus hash. This block tests whether layer decoding is driven by response consistency rather than by average response amplitude alone.

Feature	Description
b_amp_var b_amp_std b_amp_cv	Variance of response amplitudes across repeated trials of the same hash. Standard deviation of response amplitudes across repeated trials. Coefficient of variation: $CV = \frac{std}{mean}.$
b_amp_fano	It is undefined or set to missing when the mean response is too small. Fano-like factor: $\frac{variance}{mean}.$
b_amp_snr	It measures trial-by-trial dispersion relative to the mean response. Signal-to-noise ratio: $\frac{mean}{std}.$
b_trace_corr_mean	High values indicate that the mean response is large relative to trial variability. Mean Pearson correlation between all pairs of complete trial traces for the same neuron and hash. It measures similarity of trace shape across repetitions.
b_oracle_corr	Leave-one-trial-out oracle reliability. Each trial is compared with the average of the remaining trials, giving a measure of reproducible stimulus-locked response structure.

Tier C1: Behavioural modulation

Tier C1 contains 4 features describing how the neural response changes with behavioural state. These features are motivated by the fact that visual responses in mouse V1 can be modulated by locomotion, pupil size, and arousal.

Feature	Description
c1_rmi	Running Modulation Index: $RMI = \frac{resp_{running} - resp_{still}}{resp_{running} + resp_{still} + \epsilon}.$
c1_pupil_resp_slope	Values near +1 indicate stronger responses during running, values near -1 indicate stronger responses during stillness, and values near 0 indicate little modulation. Slope of an ordinary least squares regression between trial-by-trial response amplitude and trial-by-trial pupil dilation. It measures whether responses increase or decrease with pupil size.
c1_state_rel_diff	Difference in response variability between running and still trials: $std(amp \mid running) - std(amp \mid still).$
c1_arousal_gain_diff	It measures whether reliability depends on behavioural state. High-pupil modulation index, analogous to the running modulation index but comparing high-pupil and low-pupil states. It captures arousal-related response gain.

Tier D1: Stimulus descriptors

Tier D1 contains stimulus-level descriptors. Unlike A1, B and C1, these features describe the visual stimulus rather than the neuron directly. They test whether layer decoding benefits from explicit information about the visual input that evoked the response.

D1 numeric stimulus features

The numeric D1 features describe luminance, contrast, motion energy, and spatial or temporal frequency content.

Feature	Description
d_luminance_mean d_luminance_std	Mean pixel intensity of the stimulus, after normalization to the range $[0, 1]$. Standard deviation of pixel intensities. It measures visual variability or luminance dispersion.

Feature	Description
d_contrast	Contrast-like measure computed as: $\frac{\text{luminance_std}}{\text{luminance_mean}}.$
d_motion_energy	It captures relative image contrast. Mean absolute difference between consecutive frames: $\text{mean } frame(t + 1) - frame(t) .$
d_sf_low	It measures how much the stimulus changes over time. Fraction of spatial Fourier power in low spatial frequencies, approximately 0 to 0.1 cycles/pixel. Low spatial frequencies correspond to broad, coarse image structure.
d_sf_mid	Fraction of spatial Fourier power in mid spatial frequencies, approximately 0.1 to 0.3 cycles/pixel.
d_sf_high	Fraction of spatial Fourier power in high spatial frequencies, approximately 0.3 to 0.5 cycles/pixel. High spatial frequencies correspond to fine detail, edges or texture.
d_tf_low	Fraction of temporal Fourier power in low temporal frequencies, approximately 0 to 0.1 cycles/frame. It captures slowly changing stimuli.
d_tf_mid	Fraction of temporal Fourier power in mid temporal frequencies.
d_tf_high	Fraction of temporal Fourier power in high temporal frequencies. It captures rapidly changing or flickering stimuli.

C Full results plots

This appendix collects the full result panels behind the summary figures in the main text.

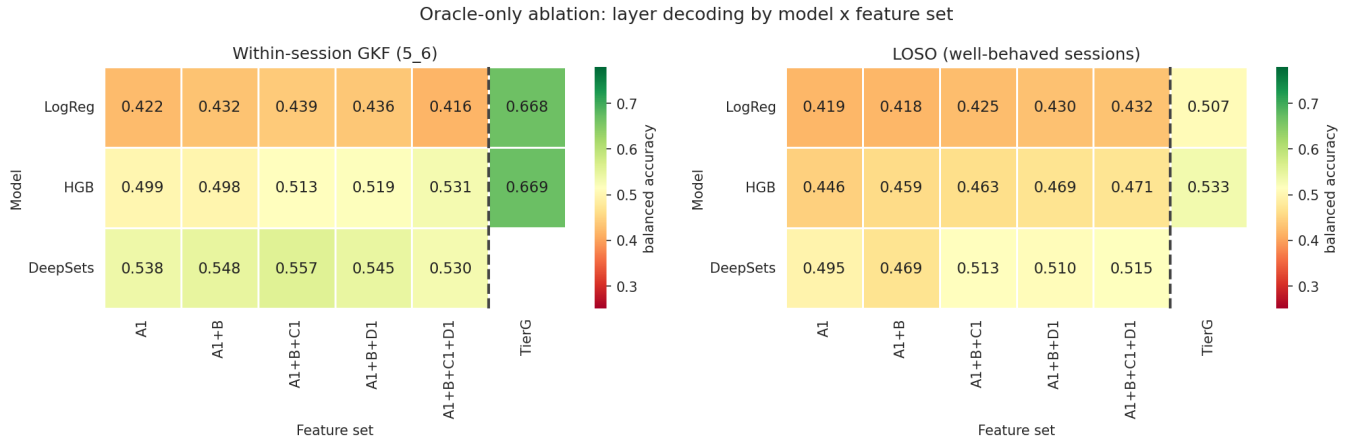


Figure 4: Tabular ablation, both protocols. Balanced accuracy for LogReg, HGB and Deep Sets across the five engineered feature stacks (A1, A1+B, A1+B+C1, A1+B+D1, A1+B+C1+D1) and the Tier G oracle fingerprint. Left: within-session GroupKFold ceiling. Right: leave-one-session-out (LOLO). The GKF→LOLO drop quantifies session shift; Deep Sets does not apply to Tier G (one row per neuron).

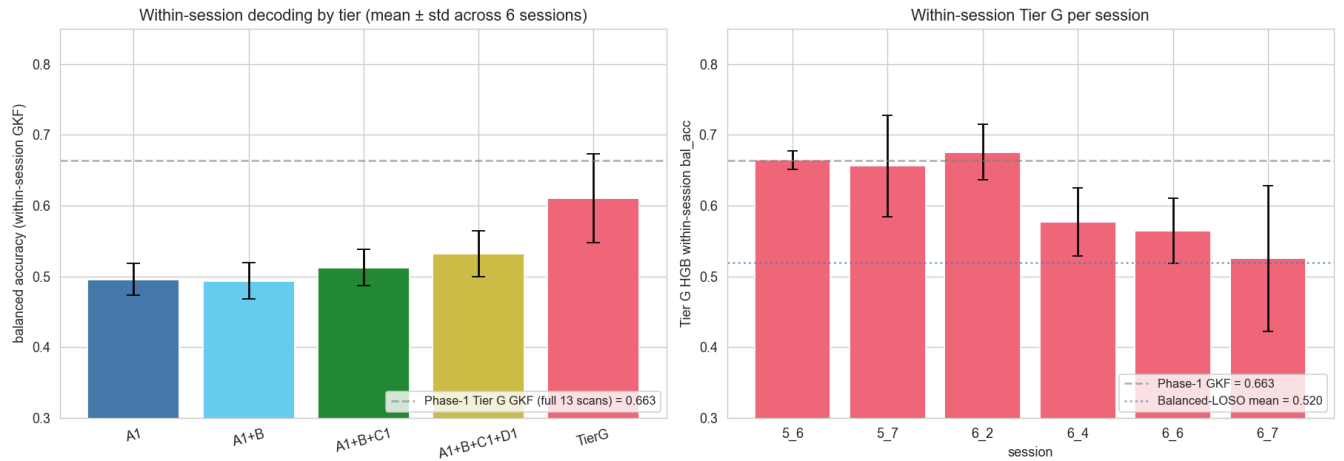


Figure 5: Within-session results. Left: within-session decoding by feature tier (mean $\pm$ std across the well-behaved sessions). Right: Tier G HGB within-session balanced accuracy per session, against the Phase-1 GKF ceiling and the balanced-LOLO mean.

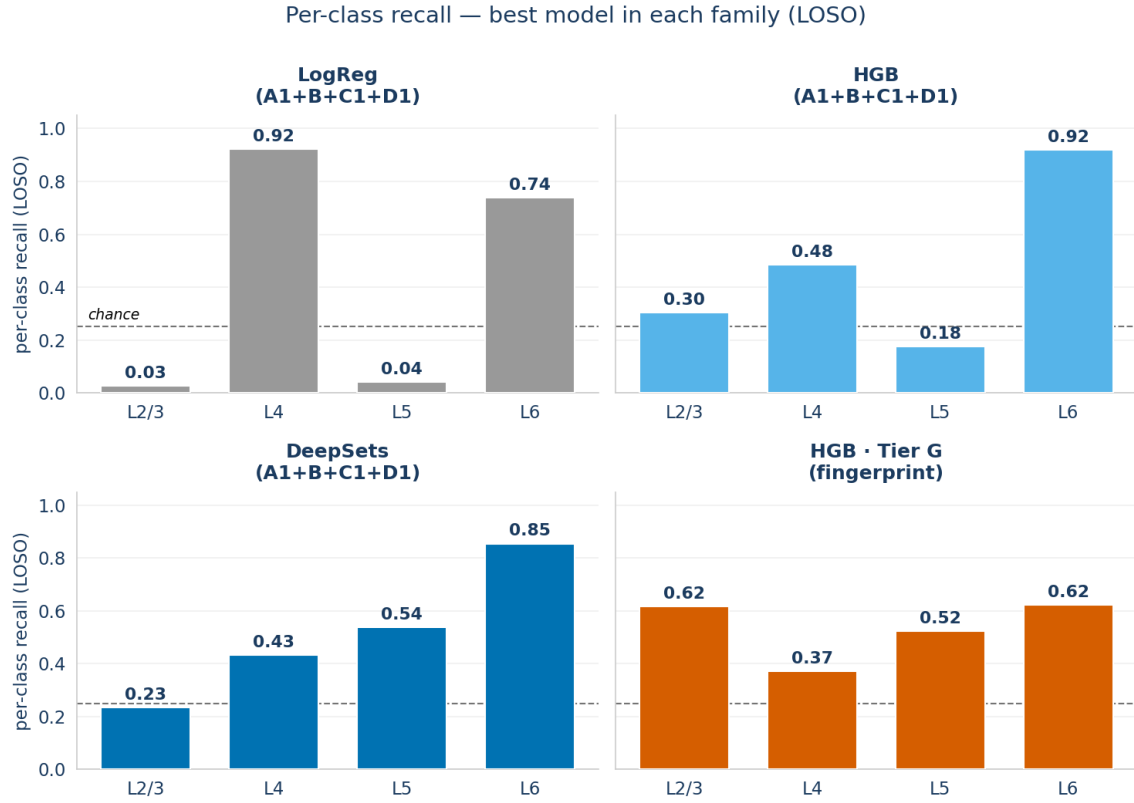


Figure 6: Per-class recall: Tier G vs. tabular. Best model in each family under LOSO. Tier G is the only representation that recalls every layer above chance; the tabular models concentrate on L4/L6 and miss L2/3 and L5.

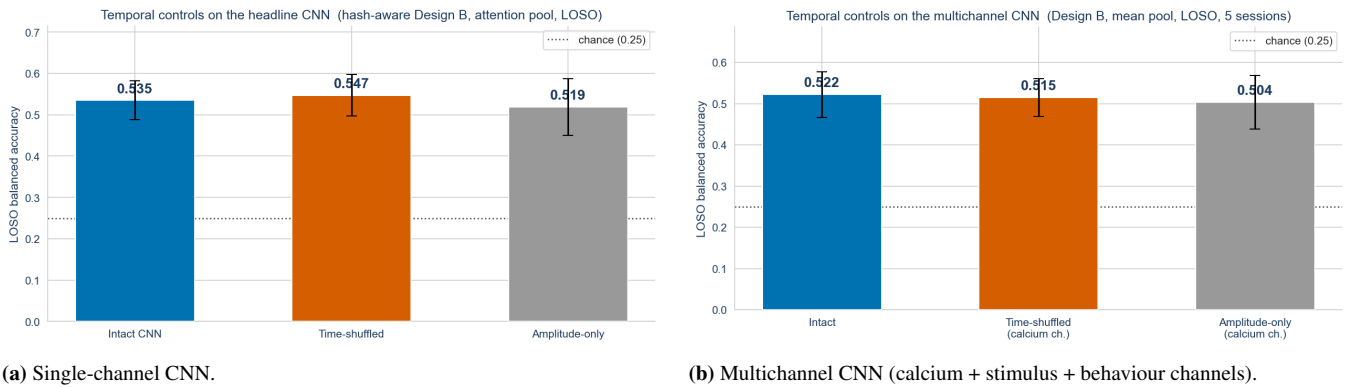


Figure 7: CNN temporal controls, LOSO. Intact vs. time-shuffled vs. amplitude-only, on both the single-channel and the multichannel CNN designs. Single-channel: intact 0.535, time-shuffled 0.547, amplitude-only 0.519. Multichannel: intact 0.522, time-shuffled 0.515, amplitude-only 0.504. In both designs the three conditions are within ~ 1 – 2 percentage points, so fine temporal order is not the source of the CNN’s layer signal.

D Full 13-session extension

The main benchmark used five balanced sessions to compare representations under identical conditions. As a broader extension, the analysis was also run on the larger 13-session population. These results test whether the qualitative conclusions persist when more data are included, but they are less directly comparable to the CNN benchmark because class balance and session composition are less controlled. In particular, three scans (9_3, 9_4, 9_6) contain only a single target layer, so their per-session balanced accuracy is degenerate and should not be averaged naively with the four-class scans.

Table 8: All-13-session LOSO results. Per-session balanced accuracy for the Tier G oracle fingerprint (HGB, z-scored) and for the engineered tabular headline (A1+B+C1+D1 HGB). Averaging over the ten multi-layer scans (the three single-layer scans 9_3, 9_4, 9_6 are excluded because balanced accuracy is degenerate when only one class is present), the mean is 0.470 for Tier G and 0.402 for the tabular headline — the same ordering as on the five-session benchmark, confirming the qualitative conclusion holds on the broader, less controlled population.

Session	n test	# classes	Tier G	Tabular
4_7	715	3	0.465	0.325
5_6	686	4	0.601	0.517
5_7	701	4	0.505	0.440
6_2	774	4	0.450	0.480
6_4	916	4	0.415	0.451
6_6	697	4	0.437	0.452
6_7	874	4	0.446	0.434
7_3	552	3	0.398	0.322
7_5	274	3	0.452	0.326
8_5	780	3	0.529	0.277
9_3	818	1	0.361	0.000
9_4	680	1	0.700	0.004
9_6	428	1	0.717	0.488
Multi-layer mean (10 scans)			0.470	0.402

E Side exploratory analyses

E.1 Stimulus-family analysis

Stimulus-family analyses were used to ask whether layer information is uniformly distributed across visual stimulus families or concentrated in particular stimulus regimes. These analyses are exploratory because splitting by family reduces the amount of data and introduces a new biological question beyond the main representation benchmark.

Within-family decoding (Table 9) shows that all three families carry layer information, with the parametric Monet2 gratings the strongest single-family reporter and the natural-movie Clip family close behind. Cross-family transfer (Table 10) collapses substantially: a model trained on one family and tested on another stays only modestly above the 0.25 chance level, and HGB trained on Clip collapses entirely to chance when transferred. This indicates that the layer signature is partly stimulus-class-specific rather than a single universal code.

Table 9: Within-family layer decoding (A1 features, within-family GroupKFold balanced accuracy).

Family	LogReg	HGB	Deep Sets
Clip	0.417	0.596	0.483
Monet2	0.485	0.614	0.530
Trippy	0.441	0.559	0.480

Table 10: Cross-family transfer (HGB, A1 features, balanced accuracy; train family in rows, test family in columns). Chance = 0.25.

Train \ Test	Clip	Monet2	Trippy
Clip	—	0.251	0.249
Monet2	0.449	—	0.422
Trippy	0.441	0.405	—

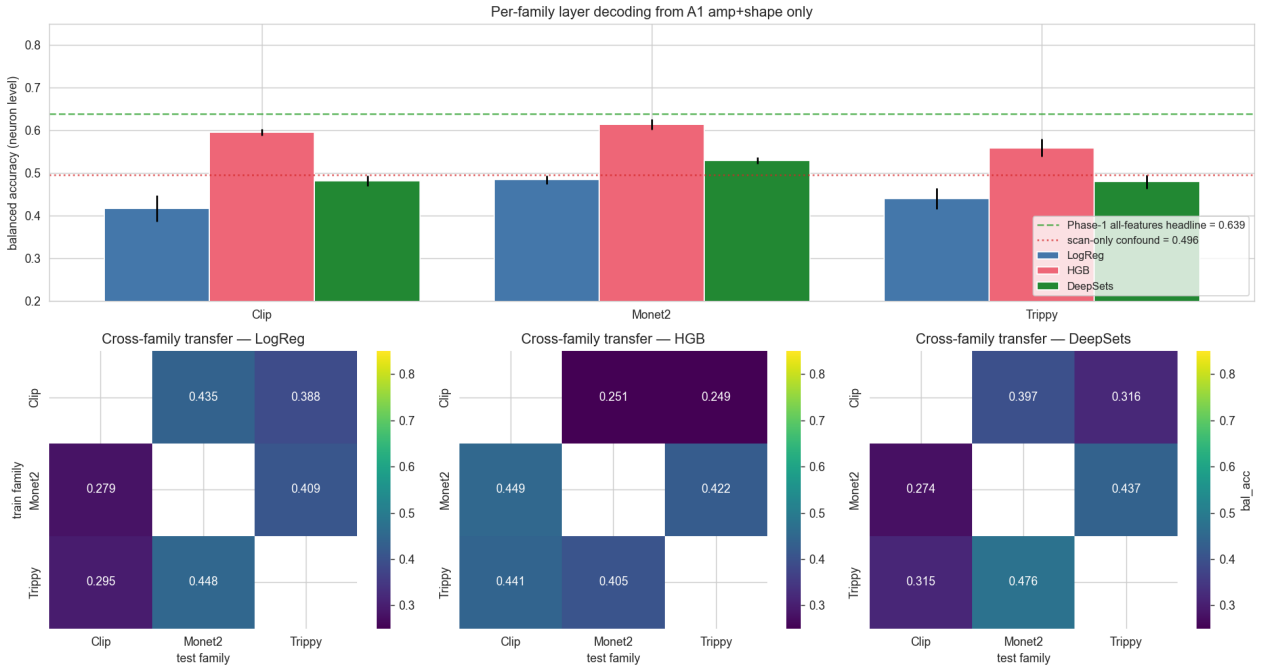


Figure 8: Per-stimulus-family layer decoding. Within-family balanced accuracy for the three stimulus families.

E.2 Excitatory subtype decoding

After layer decoding, we explored finer excitatory subtype decoding. These analyses were treated as exploratory because subtype labels are more granular and severely imbalanced, especially for rare classes such as 5P-NP and the L6 subtypes. The most biologically interpretable targets are within-layer binaries, especially L5 IT vs. ET and L6 IT vs. CT.

The within-layer binaries are weakly but consistently above chance under LOSO, and survive a cc.abs (response-quality) residualization control (Table 11). The full seven-class problem is only modestly above its $1/7 \approx 0.14$ chance level, and is dominated by the abundant 23P and 4P classes; the rare subtypes (5P-NP, the L6 subtypes) are essentially not recoverable. Subtype decoding is therefore reported as an exploratory extension and not as a main claim.

Table 11: Excitatory subtype decoding (LOSO balanced accuracy, valid scans only). “Resid.” rows residualize against the cc_abs response-quality scalar. Binary chance = 0.50; seven-class chance ≈ 0.14 .

Task	Model / protocol	Bal. acc.
L5 IT vs. ET	A1+B HGB	0.606 ± 0.094
L5 IT vs. ET	A1+B+C1+D1 HGB, resid.	0.582 ± 0.079
L6 IT vs. CT	A1+B+C1+D1 LogReg, resid.	0.570 ± 0.066
7-class subtype	A1+B+C1+D1 HGB	0.284 ± 0.069
7-class subtype	A1+B+C1+D1 HGB, resid.	0.337 ± 0.076